

TITLE PAGE

- **Problem Statement ID – Q4**
- **Problem Statement Title-** AI-Powered Behavioral Anomaly Detection for Cybersecurity
- **Theme-** AI-Powered Behavioral Anomaly Detection for Cybersecurity
- **PS Category-** Software
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AI-Powered Behavioral Anomaly Detection for Cybersecurity

Detailed Explanation of the Proposed Solution

- AI-powered Enterprise UEBA platform integrating Isolation Forest, Autoencoder, FastAPI, Streamlit, SHAP, and AI Analyst Copilot.
- Uses real-time user and entity telemetry to detect behavioral anomalies, assess risk, and automate incident investigations.

How It Addresses the Problem

- Replaces alert-centric monitoring with incident-centric behavioral intelligence.
- Detects insider threats and compromised accounts using hybrid AI models.
- Accelerates SOC investigations through automated risk scoring and AI-generated incident summaries.

Innovation & Uniqueness

- Hybrid AI engine combining Isolation Forest and Deep Autoencoder for anomaly detection.
- Behavioral Identity Engine builds adaptive user baselines for continuous risk assessment.
- AI Analyst Copilot generates MITRE ATT&CK mappings and one-click remediation scripts.

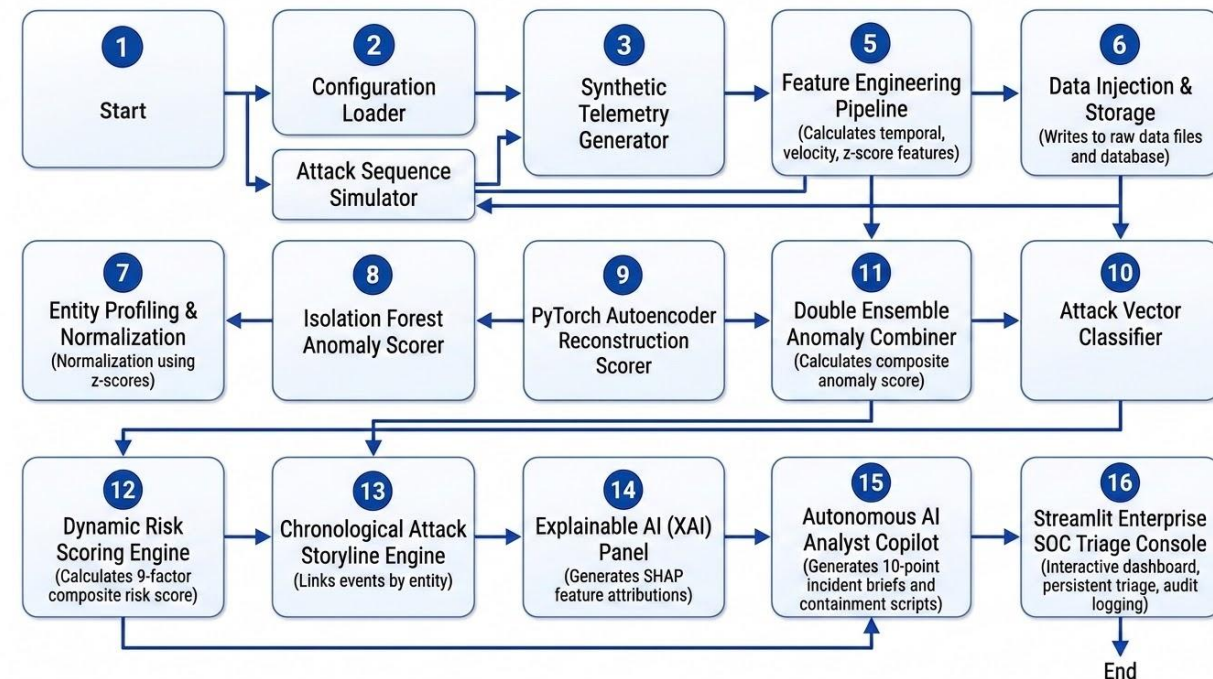
TECHNICAL APPROACH

Technologies Used

1. ML Engine: Isolation Forest, PyTorch Autoencoder
2. AI Copilot: LLaMA 3.1 (Groq), SHAP
3. Backend: Python, FastAPI
4. Dashboard: Streamlit, Plotly, Pandas

Implementation Methodology

1. Collect real-time user and entity telemetry.
2. Extract behavioral and temporal features.
3. Detect anomalies using a hybrid AI ensemble.
4. Calculate dynamic multi-factor risk scores.
5. Generate AI-powered incident summaries.
6. Recommend remediation and analyst actions.



FEASIBILITY AND VIABILITY

Analysis of the Feasibility of the Idea

- Built using production-ready open-source frameworks (PyTorch, Scikit-Learn, FastAPI, Streamlit).
- Real-time anomaly detection (<50 ms/event) without requiring GPU infrastructure.
- Integrates seamlessly with enterprise SIEM platforms and standard telemetry logs.

Potential Challenges and Risks

- Data drift causing false positives over time.
- Low-and-slow attacks bypassing anomaly thresholds.
- High-volume telemetry processing and compliance requirements.

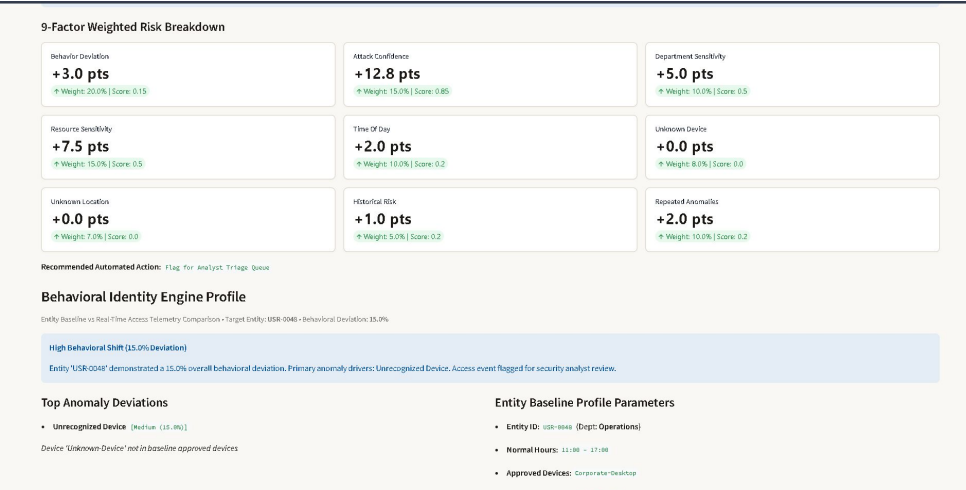
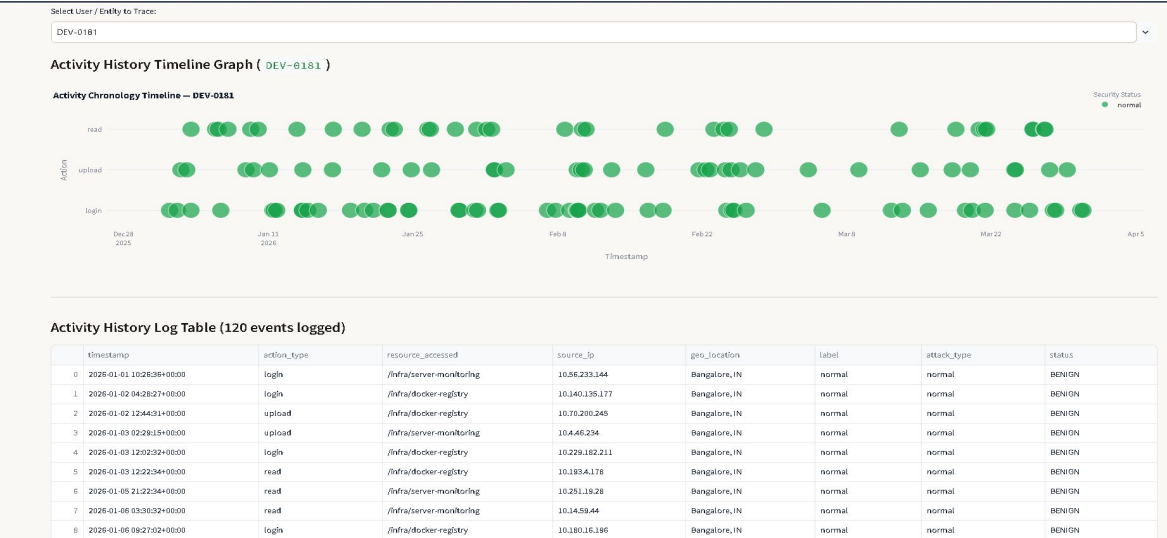
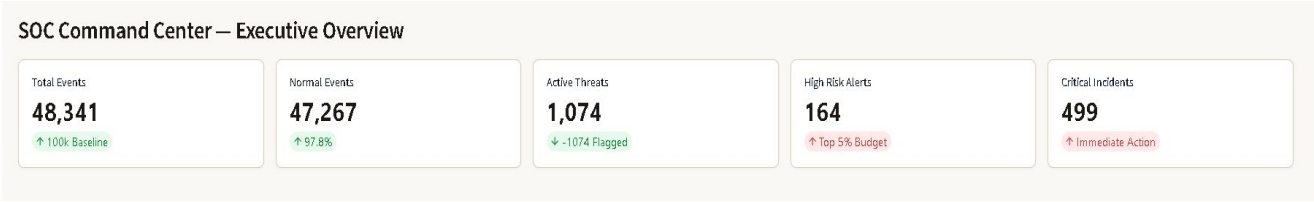
Strategies for Overcoming These Challenges

- Adaptive behavioral baselines with continuous model updates.
- Multi-factor risk scoring and attack storyline correlation.
- SHAP explainability, analyst feedback loop, and scalable microservices.

ARTIFACTS

DashBoard Screenshot

https://github.com/rajsenyadav/SentinelAI



RESEARCH AND REFERENCES

- Isolation Forest by Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou (2008) for statistical anomaly detection.
- Autoencoder-based Anomaly Detection by Masashi Sakurada and Takehisa
- Yairi (2014) for deep learning anomaly detection.
- SHAP (SHapley Additive exPlanations) by Scott M. Lundberg and Su-In Lee (2017) for explainable AI.
- MITRE ATT&CK Framework for attack tactics, techniques, and threat mapping.
- NIST Cybersecurity Framework (CSF 2.0) for cybersecurity best practices and risk management.
- NIST SP 800-61 Rev.2 – Computer Security Incident Handling Guide for incident response methodology.